SUBJECT: PHYSICS

CLASS: 10th

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DEDICATED TO

تمام تعریفیں رب لاشریک کے لئے جس نے مجھ ناچیز کو پیدا کیا۔ اور لاکھوں کروڑوں درود اس ذات پر جس نے ہماری بخشش کے لئے رو رو کر دعا مانگی۔ ان نوٹس کو امت محمدیہ کی آسانی کے لئے بنایا گیا ہے۔ ان کا ثواب رب موسیٰ و ہارون نبی آخرالزماں کی جملہ آل و اولاد اور امت کو عطا کرے۔ آمین

راجہ حمزہ خان

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انتباہ

یہ نوٹس مفاد عامہ کے لئے بنائے گئے ہیں۔ ان کی اشاعت باقاعدہ طور پر ڈی سی آفس میں رجسٹرڈ ہے۔ ان کو بیچنا یا کسی بھی طور پر مالی فائدے کے لئے آگے بھیجنا قانوناً جرم ہے۔ ایسا کرنے والوں کے خلاف قانونی کاروائی کی جائے گی۔

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PHYSICS
MOST IMPORTANT SHORT QUESTIONS
CHAPTER NO 1
SIMPLE HARMONIC MOTION AND WAVES

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Q) Differ between periodic motion, vibratory motion, time period and frequency?
ANS)

PERIODIC MOTION	VIBRATORY MOTION	TIME PERIOD	FREQUENCY
Any motion which repeats itself after a fixed interval of time is known as periodic motion.	The back and forth motion of the body over the same path is known as vibratory motion.	The time taken by a body to complete one vibration is called time period. Its SI unit is second. :- T=1/F	The number of vibration made by the vibrating body in one second is called frequency. Its SI unit is Hertz. :- F=1/T

Q) Differ between amplitude, wavelength, wave and wave motion?
ANS)

AMPLITUDE	WAVELENGTH	WAVE	WAVE MOTION
The maximum distance of a vibrating body from its mean position is called as amplitude.	The distance between two consecutive crests or troughs is called wavelength.	Wave is an oscillation accompanied by the transfer of energy that travels through a medium e.g sound waves.	Wave mption is a vibratory disturbance travelling through a medium in which energy transferred from one point to another when there is no direct contact between them.

Q) Differ between electromagnetic, transverse, mechanical and longitudinal waves?

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ANS)

ELECTROMAGNETIC WAVES	TRANSVERSE WAVES	MECHANICAL WAVES	LONGITUDNAL WAVES
The wave which don't require a medium to propagate are called electro-magnetic waves. E.g radio waves. Etc	The waves in which particles of medium vibrate perpendicular to the direction of wave is called transverse waves. E.g waves in slinky. Etc	The type of wave which need a medium to propagate is called mechanical waves. E.g waves produced in stretched wire. Etc	The waves in which the particles of medium vibrate parallel to the direction of propagation of waves is called longitudinal waves. E.g sound waves. Etc

Q) Define cycle? Differ between compression and rarefaction?

ANS) A crest and its adjacent trough together forms a cycle which is also equal to wavelength.

COMPRESSION	RAREFACTION
In longitudinal waves, the regions of higher pressure where the particles are close to each other is called compression.	In longitudinal waves, the regions of lower pressure where the particles are spaced apart is known as rarefaction.

Q) Define wave front? How energy can be transmitted in vacuum?

ANS) Wave front is the surface in the path of the wave over which all the particles have the same phase of vibration.

Energy can be transmitted through vacuum in the form of electromagnetic waves.

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Q:- Are sound waves are longitudnal or transverse in nature?

Q: What is the period of a pendulum whose length is 4cm.

Ans: Sound waves are longitudnal in nature because the particles of medium vibrate parallel to the direction of propagation of waves.

⟹ Length = 4cm, $g = 9.8 m/s^{-2}$

As we know that

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T = 2 \times \frac{22}{7}\left(\sqrt{\frac{4}{9.8}}\right)$$

$$T = 2 \times 3.1428 \times (\sqrt{0.4081})$$

$$T = 2 \times 3.1428 \times 0.6388$$

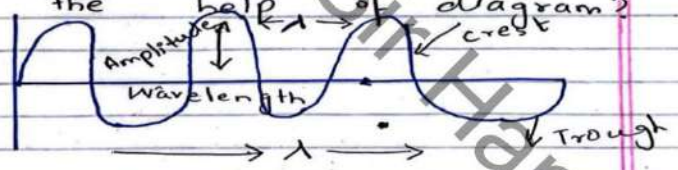
$$T = 2 \times 2.007$$

$$T = 4.015$$

$$T = 4s$$

Q2: Explain the term amplitude with the help of diagram?

Ans:



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CHAPTER NO 2
SOUNDS

Q) How sound is produced? Write three things which are necessary for the production of sounds?

ANS) Sound is produced by vibrations. To vibrate a body, some mechanical energy is required. The mechanical energy is converted into sound energy. Three things necessary for the production of sound are:

Vibrating body

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Transmitting medium
Hearing device

Q) Differ between loudness, intensity, quality and pitch of sound?
ANS)

LOUDNESS	INTENSITY	QUALITY	PITCH
The characteristic of sound by which a loud sound and a faint sound can be distinguished is called loudness.	Energy passing per second through a unit area held perpendicular to the direction of propagation of sound waves is called intensity.	The property of sound by which two sounds of same loudness and pitch are distinguished from each other is called quality of sound.	The characteristic of sound by means of which a shrill sound can be distinguished from a grove sound is called pitch of sound.

Q) Differ between musical sound, noise, speed of sound and audible frequency range?
ANS)

MUSICAL SOUNDS	NOISE	SPEED OF SOUND	AUDIBLE FREQUENCY RANGE
Sound which has pleasant effect on our ears is called musical sound. E.g sounds by playing piano. Etc	Sound which has unpleasant effect on our ears is called noise. E.g sounds of drill machine, vehicles.	The distance travelled per unit time by a sound as it propagated through a medium	Those sounds whose frequency is between 20Hz to 20KHz are called audible frequency range.

Q) Define ultrasound? Write its two applications? What would happen if sound could travel through vacuum?
ANS) The sounds having frequency greater than 20000 Hz are called ultrasound.
APPLICATION OF ULTRASOUND:
1 Ultrasound is used to investigate defect in internal organs of human body.
2 Ultrasound is used for finding the level of liquid in a metal tank. (Without opening)

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If slund could travel through vacuum, it would be regarded as Electromagnetic in nature than longitudinal.

Q) In which medium air or water, an echo is heard much sooner. Why? Why is echo weaker than original sound? Differ between ultrasonic and infrasonic?

ANS) Echo can be heard much sooner through liquid because speed of sound in liquid, being dense, is higher in comparison to air.

As the sound travels, some sound of original one is absorbed by surrounding medium and by surface it reflects from. As a result, echo is weaker than original sound.

ULTRASONIC	INFRASONIC
Sound having frequency more than 20 KHz is called ultrasonic.	Sound having frequency less than 20 Hz is called infrasonic.

CHAPTER NO 3
GEOMETRICAL OPTICS

Q) Define reflection? Write its laws also?

ANS) The process of bouncing back of the light rays, which fall on the surface of an object is called reflection of light.

FIRST LAW OF REFLECTION:

The incidence ray, the reflected ray and the normal all lie in the same plane.

SECOND LAW OF REFLECTION:

The angle of incidence is equal to the angle of reflection.

Q) Differ between spherical mirror, spherical mirror formula, concave mirror and convex mirror?

ANS)

SPHERICAL MIRROR	SPHERICAL MIRROR FORMULA	CANCAVE MIRROR	CONVEX MIRROR
A section of hollow sphere	The relation between the	The spherical mirror in which the	The spherical mirror in which the

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with its inner or outer surface is polished to reflect light is called a spherical mirror. It has two types: Convex, concave	image distance, object distance and focal length of spherical mirror is called spherical mirror formula. **1/f = 1/p + 1/q**	reflection of light take place at the concave surface is called concave mirror. E.g inner surface of spoon. Etc	reflection of light take place at the convex surface is called convex mirror. E.g outer surface of spoon. Etc

Q) Differ between focal length, center of curvature, Snell's law and refractive index?
ANS)

FOCAL LENGTH	CENTER OF CURVATURE	SNELL'S LAW	REFRACTIVE INDEX
Focal length is the distance between the center of spherical lens and its focus.	The center of hollow sphere of which mirror is a part is called center of curvature.	For a ray incident on the interface of two medium, the ratio between the sine of angle of incidence of the first medium to the sine of angle of refraction of the second medium is contant.	Refractive index is the ratio of speed of light in vacuum to the speed of light in any other medium. **n=c/v**

Q) Define refraction? Also write its laws?
ANS) The bending of light ray when it passes from one transparent medium to another is called refraction of light.
FIRST LAW OF REFRACTION:
The incidence ray, refracted ray and normal at the point of incidence all lie in same plane.
SECOND LAW OF REFRACTION:
When ray of light passes from one medium to another, the ratio of sine of angle of incidence to sine of angle of refraction is constant.

Q) Differ between critical angle, total internal reflection, linear magnification and power of lens?
ANS)

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CRITICAL ANGLE	TOTAL INTERNAL REFLECTION	LINEAR MAGNIFICATION	POWER OF A LENS
The angle of incidence in the denser medium for which the corresponding angle of refraction is 90 in the rare medium is called critical angle.	When angle of incidence becomes greater than critical angle, instead of refraction, light is totally reflected back in the denser medium is called total internal reflection.	It is the ratio of height or distance of image to the height or distance of object. **M= q/p** It is denoted by M.	The degree of divergence and convergence of light ray is expressed in terms of power of lens.v **P=1/f** Its SI unit is diopter.

Q) Differ between magnifying power and resolving power?
ANS)

MAGNIFYING POWER	RESOLVING POWER
Ratio of angle subtended at the eye by the image through an optional instrument to the angle subtended at the eye by the object when seen directly is called magnifying power.	Ability of an optional instrument to form distinctly separate image of rwo objects that are very close to each other is known as resolving power.

Q) Define eye? Differ between myopia and hyper-metropia?
ANS) Eye is a natural optical instrument present in the sockets of our skull having a diameter of 2.5 cm. It functions just like a camera.

MYOPIA	HYPER-METROPIA
It is an eye defecy in which close objects appear clearly but distant objects appears blurry.	It is an eye defect in which distant objects seen clearly but close objects seen blurry.

Q) Why can a near sighted person not focus clearly on distant objects? Can we observe total internal reflection on air water interface of pond?

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ANS) A near sighted person cannot focus clearly on a distant object because the image of a distant object is formed in front of the retina and not at the retina itself. No, we cannot observe total internal reflection on the air water interface of the pond because total internal reflection cannot occur when light travels from light to denser medium.

Q) At what place one should stand in front of a concave mirror to hide his/ her image? Why is negative sign associated with virtual image?
ANS) When the object is placed at the focus point of concave mirror no image is formed.
The negative sign associated with virtual image indicates that focal length is negative.

Q) How can you distinguished between a real and virtual image?
ANS)

REAL IMAGE	VIRTUAL IMAGE
When rays of light converge after reflection fron the mirror, the image formed is considered as real.	When rays of light diverge after reflection from the mirror, the image formed is considered as virtual.

CHAPTER NO 4
ELECTROSTATIC

Q) Differ between electric charge, electrostatic induction, electric field and electric field intensity?
ANS)

ELECTRIC CHARGE	ELECTROSTATIC INDUCTION	ELECTRIC FIELD	ELECTRIC FIELD INTENSITY
It is a physical property of matter that causes it to experience a force when placed in an electric field. Its SI unit is Coulomb. It is denoted by q.	The redistribution of charges on a neutral body due to the presence of a charged body nearby it, is called electrostatic induction.	Electric field is the strength of electric charge at a given point.	It is the electrostatic force per unit charge at a point. Its unit is N/C.

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Q) Differ between electrostatic potential and potential difference?
ANS)

ELECTROSTATIC POTENTIAL	POTENTIAL DIFFERENCE
Electrostatic potential is work done on a charge to move it from infinity to a point in the electric field.	It is the work done on a charge to move it from a point to another against the electric field.

Q) Define dielectrics? Also prove that 1 volt= 1 Joule / 1 Coulomb?
ANS) Dielectrics can be air or sheets of same insulating material.
1 volt = 1 Joule / 1 Coulomb
Electric potential between two points will be 1V if 1 J work is done in moving a charge of 1C from 2 point to another against the electric field.
V= W/q.
1 volt=1 Joule/1 Coulomb, 1V=1J/1C

Q) How a charge is developed on a neutral body? How an insulator can be changed by electrostatic induction? Why a metal chain is hung at the rear end of fuel tankers?
ANS) A charge is developed on neutral body by bringing a charged body closed to it. The neutral body may experience attraction or repulsion due to opposite or same charges.
By bringing the insulator near the charge body, all the charges on an insulator will redistribute.
Because to discharge any build up of static charge that may build up duri the journey so that when the fuel is being loaded or off-loaded the charge won't set the fuel alight.

CHAPTER NO 5
CURRENT ELECTRICITY

Q) Differ between electric current, conventional current, potential difference and emf with SI unit and form?
ANS)

ELECTRIC CURRENT	CONVENTIONAL CURRENT	POTENTIAL DIFFERENCE	EMF

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Electric current is the rate of flow of charges through a cross-sectional area of a conductor. I= q/ t Its SI unit is Ampere. 1A= 1C/s	Conventional current flows due to the flow of positive charges from a point of higher to a point of lower potential.	Potential difference is the difference in electric potential between two points in a circuit. V=W/q . Its SI unit is Volt.	emf is the voltage between the terminals of a cell or a battery in a circuit. emf= energy/charge. Its SI unit is volt.

Q) Define OHM's law with SI unit and form? Differ between resistance and electric power with SI unit and form?

ANS) The current passing through a conductor is directly proportional to the potential difference across its ends, if the temperature and physical state remain constant. **V=IR**

Its SI unit is resistance.

RESISTANCE	**ELECTRIC POWER**
It is the property of a conductor by which it opposes the flow of current passing through it. Its SI unit is OHM. R=V/I	It is the amount of energy supplied by the electric current in unit time. Its SI unit is Watt. P=VI

Q) Differ between Series and parallel combination of resistors and write the equation for equivalent resistors?

ANS)

SERIES COMBINATION	**PARALLEL COMBINATION**
Series combination of resistors is the combination of resistors in end to end arrangements such that some current passes through each resistor. Their equivalent resistance will be: Re= R1+R2+R3+------+Rn	Parallel combination of resistors is the side by side arrangements of resistors with their ends joined at common points such that sum of potential difference is same across each resistor. Their equivalent resistance is: 1/Re= 1/R1+1/R2+1/R3+-------+1/Rn

Q) Differ between conductor, insulator, Electrical energy and Joules law?

ANS)

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CONDUCTOR	INSULATOR	ELECTRICAL ENERGY	JOULES LAW
Conductors are the materials which allow electricity to pass through them. E.g silver. Etc	Insulators are the materials which don't allow electricity to pass through them. E.g plastic. Etc	Electrical energy is the amount of energy generated in a resistance due to flow of charges. W= QV	It is the ratio of product of potential difference square and time to tge resistance of circuit. W=V^2t/R

Q) Define kilowatt-hour? Prove 1KWh= 3.6 MJ? Differ between direct and alternating current?
ANS) It is the amount of energy delivered by a power of one kilowatt in one hour.
1 kilowatt-hour = (1000W)(3600s)
1 KWh = 3600000 Ws
1 KWh = 3.6 MJ (Ws= M) (10^6= M)

DIRECT CURRENT	ALTERNATING CURRENT
Direct current does not changes direction and flow in only one direction. E.g battery, cell. Etc	Alternating current changes direction after equal interval of time. E.g current from AC generator. Etc

Q) On what factors the resistance of a conductor depends? Why is it dangerous to switch on a light bulb with wet hands?
ANS) The resistance of a conductor depends upon the length, cross sectional area and nature of material of a conductor.
Because wet skin has a very low resistance of a few hundred ohm's whereas dry human skin has a resistance of 100000 ohm's.

Q) How does fuse wire prevent the danger of short circuit? Wht is conventional current considered as an equivalent to electronic current?
ANS) A fuse is used in live wire to prevent short circuits. The fuse wire geys heated up and it meltscand creates a gap in the live wire and current cannot flow through it and is saved from danger.

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Conventional current produces the same effect as current flowing from negative and positive terminal due to flow of negative charges. So it can be considered equivalent to electronic current.

CHAPTER NO 6
ELECTROMAGNETISM

Q) Differ between magnetic field, magnetic flux, solenoid and Electromagnet?
ANS)

MAGNETIC FIELD	MAGNETIC FLUX	SOLENOID	ELECTROMAGNET
The region surrounding a current carrying conductor is called magnetic field.	The number of magnetic field lines passing through an area is called its magnetic flux.	It is a coil with many circular turns of indulated copper wire wrapped closely in the shape of cylinder.	Electromagnet is the type of magnet in which the magnetic field is produced by an electric current.

Q) Define Tesla and Right Hand Rule?
ANS) Tesla: The magnetic field strength will be 1 tesla when a conductor of length 1 m carrying a current of 1A perpendicular to field experiences a force of 1N.
1T= 1N/A/m
Right hand rule: If we hold the current carrying wire in the right hand, such that the thumb is directed in direction of current, the curled fingers will show the direction of magnetic field.

Q) Define faraday law of electromagnetic induction? Differ between lenz law and mutual induction?
ANS) Faraday's law of electromagnetic induction is stated as, The emf induced in a coil depends upon the number of magnetic field lines through the coil.

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LENZ LAW	MUTUAL INDUCTION
It states that the direction of induced current is opposite to the current source.	Mutual induction is the generation of an emf in a coil due to change in current in a nearby coil.

Q) Define transformer? Differ between step-up ans step-down transformer?
ANS) It is an electrical device which increases or decreases the value of alternating current.

STEP-UP TRANSFORMER	STEP-DOWN TRANSFORMER
If secondary voltage is larger than the primary voltage, the transformer will be a step-up transformer. Ns>Np So, Vs>Vp	If secondary voltage is less than primary voltage, the transformer will be a step-down transformer. Ns<Np So, Vs<Vp

Q) What is the function of carbon brushes in Ac generator? Differ between primary and secondary coil?
ANS) In Ac generator, two carbon brushes are attached to the slip rings. These carbon brushes conduct the current between these brushes.

PRIMARY COIL	SECONDARY COIL
The coil in which the change in current causes induced current in another coil is called primary coil.	The coil in which current is induced is called secondary coil.

CHAPTER NO 7
INTRODUCTORY ELECTRONICS

Q) Differ between thermionic emission, electronic, analogue and digital electronics?
ANS)

THERMIONIC EMISSION	ELECTRONICS	ANALOGUE ELECTRONICS	DIGITAL ELECTIONS

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The process by which electrons are emitted by a hot metal surface is called ther-mionic emission.	The branch of physics which deals with the behavior of electrons using different devices for useful purpose	It is a branch of electronics in which circuits work with a continuously variable signal.	It is a branch of electronics in which circuit use a digital signal based on binary numbers (0,1).

Q) Define electron gun? Differ between NAND and NOT gate?
ANS) An electron gun is a device which is used to produce a narrow collimated beam of electrons and studybthe behavior of an electron beam.

NAND GATE	NOT GATE
A NAND gate is formed by combining a NOT and AND gate. The NAND gate is expressed as X=A.B.	NOT gate is a logic gate which gives value one when input is zero and zero when input is one.

Q) Define logic gates? Differ between AND and OR gate?
ANS) Logic gate is an electronic circuit having one or more input but only one output.

AND GATE	OR GATE
AND gate is a lpgic gate which gives the value of one if and only if all the input values are one.	OR gate is a logic gate which gives value one if any of the input value is one.

CHAPTER NO 8
INFORMATION AND COMMUNICATION TECHNOLOGY

Q) Differ between information technology, telecommunication, E-mail and telephone? How is data stored on audio cassette?
ANS)

INFORMATION TECHNOLOGY	TELECOMMUNICATION	E-MAIL	TELEPHONE

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It is the study and use of system for storing, retrieving and sending information.	It deals with the transmission and reception of information over long distances.	E-mail is a method for sending and receiving messages through internet.	Telephone is a device which allows two-way communication of sound over long distances.

A cassette stores the sound on a magnetic tape that is wound around the two reels in the cassette.

Q) Define internet and write its four uses?

ANS) Internet is a global network system. It is a very useful outcome of information technology. It enable us to gather information and communicate with any person on globe.

USES OF INTERNET:

Used to check E-mails.
Used for research.
Updating software.
Entertainment, shopping, data sharing. Etc

CHAPTER NO 9
RADIOACTIVITY

Q) Define atom? Differ between atomic number, mass number, isotopes, radioactivity?

ANS) The basic unit of matter or the building block of structure of an element is called atom.

ATOMIC NUMBER	**MASS NUMBER**	**ISOTOPES**	**RADIOACTIVITY**
The number of protons in the nucleus of an atom is called atomic number.	The sum of number of protons and neutrons present in the nucleus of an atom is called mass number.	Atom of the same element having same atomic number but different atomic masses are called isotopes.	The spontaneous emission of radiations from atoms of elements having atomic number greater than 82 is called radioactivity.

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Q) Differ between nuclear decay, half life, natural radioactivity and artificial radioactivity?
ANS)

NUCLEAR DECAY	HALF LIFE	NATURAL RADIOACTIVITY	ARTIFICIAL RADIOACTIVITY
Nuclear decay occurs when a heavy unstable nuclide changes into a light and relatively stable nuclide.	The time taken by half of the atoms of a sample of radioactive substance to decay is called half life.	The spontaneous emission of radiations from atom of elements having atomic number greater than 82 is called natural radioactivity.	The spontaneous emission of radiations from stable isotopes by bombardment of high energy radiations is called artificial radioactivity.

Q) Why there is no net charge on an atom? Why do isotopes have different number of neutrons?
ANS) An atom contains equal number of protons and electrons. Therefore, there is no net charge on an atom.
Isotopes are atoms of an element having same atomic number but different atomic masses. So, the number of neutrons would be different for every isotope of the same element.
Number of neutrons= A-Z

Q) Define radioisotopes? Why do gamma rays pass straight through an electric field?
ANS) The artificially produced radioactive elements are called radioisotopes.
6C^14.
Gamma rays pass straight through an electric field because they carry no charge.

Q) Differ between fission and fusion reactions?
ANS)

FISSION REACTION	FUSION REACTION
The spitting of heavy nucleus into two	Merging of rwo small nuclei to form a

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small nucleus with release of energy and electrons is called fission reaction.	heavy nucleus with the release of energy is called fusion reactions.

Q) Write two characteristics of alpha, beta and gamma rays?
ANS)
ALPHA RAYS:
Alpha particles are helium nuclei.
They effect the photographic plate.
BETA RAYS:
A typical source of beta particle is potassium-40.
They effect the photographic plates much greater than effect of alpha rays.
GAMMA RAYS:
A typical source of gamma rays is cobalt-60.
They cause much higher effect on photographic plate that alpha and beta rays.

MOST PROBABLE LONG QUESTION PAIRING SCHEME

LONG NUMBER 1

CHAPTER NUMBER	***CHAPTER NAME***
CHAPTER NO 1	***SIMPLE HARMONIC MOTION AND WAVES***
CHAPTER NO 3 (NUMERICAL)	***GEOMETRICAL OPTICS***

LONG NUMBER 2

CHAPTER NUMBER	***CHAPTER NAME***
CHAPTER NO 4	***ELECTROSTATIC***
CHAPTER NO 5 (NUMERICAL)	***CURRENT ELECTRICITY***

LONG NUMBER 3

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CHAPTER NUMBER	CHAPTER NAME
CHAPTER NO 8	INFORMATION AND COMMUNICATION TECHNOLOGY
CHAPTER NO 9 (NUMERICAL)	RADIOACTIVITY

MOST IMPORTANT LONG QUESTIONS

CHAPTER NO 1 (SIMPLE HARMONIC MOTION AND WAVES):
Simple harmonic motion*
Simple pendulum with example*
Ripple tank with properties
V=f(lemda)

CHAPTER NO 3 (GEOMETRICAL OPTICS):
Examples: 1,2,5,6,7
Numerical: 1,3,4,5,6,8

CHQPTER NO 4 (ELECTROSTATIC):
Electrostatic induction*
Coulombs law
Series and parallel combination of capacitor*

CHAPTER NO 5 (CURRENT ELECTRICITY):
Example: 1,2,8,9
Numerical: 1,2,3,4,5,7

CHAPTER NO 8 (INFORMATION AND COMMUNICATION TECHNOLOGY):
Transmission of information
Communication devices*
Handling information with uses of internet*

CHAPTER NO 9 (RADIOACTIVITY):
Numerical: 1,2,3,4,5

(*) Represents most important long questions.

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